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ORIGINAL TEACHING IDEAS—UNIT

Using children's literature and public libraries to prepare STEM students to communicate technical knowledge

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This unit activity describes a series of lessons designed to enhance STEM students' technical communication abilities. The lessons culminate in two separate yet interconnected projects: (1) the development of a children's book explaining a complex topic and (2) presentations of those books to members of the local community. Students learn the characteristics defining communication for problem solving and are allowed to apply that knowledge across multiple modalities, focusing primarily on the interplay between audience analysis and the simplification of scientific concepts.

Courses: *Technical Writing and Communication, Advanced Public Speaking.*

Objectives: *By the end of this activity, students should be able to (1) translate complex technical knowledge for the general public effectively, (2) conduct audience analysis, and (3) adapt messages based on the identified objectives of the communicative situation.*

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Introduction and rationale

Communication plays a vital role in helping the public reach a better understanding of science and technology (Bromme & Goldman, 2014; Bullock et al., 2019). Specifically, communication scholars understand the nuances surrounding language, instruction, audiences, and other aspects of the rhetorical situation that influence how messages are conveyed effectively. As a result, we are in a unique position to help STEM students develop their scientific communication competencies. This often means helping students understand the importance of translating their specific technical knowledge in ways that will be motivating and relevant to others without similar expertise (Frey, 2021).

However, it is difficult to find ways to connect students with nonscientists with whom they can practice these skills, despite the popularity of this approach in scientific education curricula (Brownell et al., 2013). It often takes creative connections that allow students to expand their capabilities. To illustrate, Zaelzer (2020) explored collaboration between neuroscience and art and design students to expand the efficacy of scientists' communication abilities. Providing students with the skills to recognize and adapt to their audiences holds immense utility value in that students will develop the ability to transmit their knowledge in organized and comprehensible ways. As such, this unit

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activity chronicles an attempt to train undergraduate students in a traditional face-to-face classroom to adapt their knowledge of complex topics for public consumption. Students learn key aspects of audience analysis and technical communication by writing and designing children's books that they eventually present to the public in coordination with the local public library.

The activity

Students first choose a scientific or technological concept that interests them (e.g. nuclear fission, artificial intelligence). They then create children's books explaining the concept to adolescents in the community. For my iteration of this activity, students create books via Microsoft PowerPoint. I require books to be in an electronic format and accompanied by narration from the author. However, individual instructors can make their own specific design choices.

Furthermore, students present their books, alongside a brief activity, during a special session at the local public library. Library partnerships are established prior to the start of the semester by directly contacting community liaisons. In my case, I emailed the library directly and presented them with an overview of the general assignment and learning outcomes. After making the initial connection, I collaborated with liaisons to establish relevant dates and times, as well as the structure of the event. Given that the community partner has extensive knowledge of their regular audience, partnerships should be formed conjointly. For example, the decision to accompany students' books with an illustrative activity came directly from librarians with extensive knowledge of their audience's needs.

Students have roughly one month to complete the assignment, with the library presentations of their books serving as the culmination of the project. Throughout the unit, faculty should review concepts on the rhetorical situation, technical communication, library story times, and effective speech delivery.

Day 1: The rhetorical situation

To begin, I review the foundational concepts of Bitzer's (1968) rhetorical situation: exigence, audience, and constraints. The rhetorical situation is also an important framework for understanding how technical and scientific communication is oriented as a problem-solving approach for specific audiences. Content concerning the rhetorical situation can be presented as a lecture, activity, or impromptu speaking activity (see Cooper, 2019). For example, McHendry Jr et al. (2018) provided steps for conducting an analysis of John Oliver's rants on his show *Last Week Tonight* that can be adapted into a larger discussion about the rhetorical situation. As the authors stated, understanding Oliver's persuasive goal "depends on genre (here a comedy show) and the audience (HBO subscribers are generally affluent)" (p. 67). Moreover, Oliver often tackles complex topics such as FCC regulations, cryptocurrency, and genome editing, aligning with students' topic choices. Summatively, students must first recognize the importance of adapting their communication based on the knowledge and background of their intended audience.

Day 2: Technical communication

The next lesson explains the characteristics of technical communication, or communication focused on problem solving. This includes emphasizing the importance of clear, accurate, accessible, and concise writing that uses design elements to highlight key information; activities and lectures should stress the objectivity surrounding technical communication. To illustrate, I contrast technical writing with more creative, expressive, or expository writing to show their fundamental differences. For example, I often highlight distinctions by having students outline the technical specifications of an object of their choice (e.g. how to manufacture a shoe; see Gerson, 2015, p. 5). Open-access resources—such as those by Last (2019) and Tijerina et al. (2017)—can also complement a lecture at no cost to students. Last (2019) developed a series of case studies that demonstrate the importance of explaining technical details and putting information in places where an audience is less likely to miss it (i.e. leading with key points and providing supporting information later).

Day 3: Application

To help students apply knowledge, I have them find, evaluate, or write technical manuals that provide instructions for a product or service. These directions, or other types of technical specifications, use the imperative mood, showcasing the directness and frankness that accompanies a problem-solving approach. Students should recognize how the features of technical writing complement their understanding of the audience to dictate their communication. Other useful activities have involved taking students' previous writings and revising them according to the new stylistic principles discussed in class. This could include eliminating instances of passive voice; rewriting prepositional phrases; avoiding redundant wording; creating explicit topic sentences; changing long words to short words; adding specificity, using graphics, white space, headings, lists, or other design features to control the accessibility of content; visualizing ideas graphically instead of using paragraphs; omitting superfluous information; or consolidating text into smaller sections. In any case, applying new knowledge to these deliverables gives students extended practice before eventually using the skills to build their children's books.

Day 4: Delivery

In-class content and assigned readings help students understand the principles associated with effective oral presentations. In most cases, this content can be delivered via lecture or practical activities, as documented frequently throughout the communication pedagogy literature. As an example, I frequently use the activity described by Punyanunt-Carter and Arias (2018), where students given impromptu speeches (e.g. describing a favorite tv show or someone they would like to meet) while putting explicit emphasis on *inappropriate* speaking behaviors (e.g. bad posture, no eye contact, emphasizing mistakes). Students enjoy the high-energy nature of the activity, and ensuing discussions focus on what constitutes effective delivery across contexts.

For a different perspective, I also invite an expert in public libraries and storytimes to speak with my class. Their knowledge paints a more comprehensive picture of the group who might potentially attend students' library presentations by discussing how to delivery library content appropriately to younger audiences, components of effective storytimes, and why public library events are critical for community development. Although such specializations are not available at all institutions, similar resources exist in the scholarly literature (e.g. Cahill et al., 2020) or can be facilitated through connections with the previously established community partners.

Days 5–6: Workdays

The week prior to the library presentations is largely spent in consultation with students. Specifically, strategic workdays help with the design elements of the books. I spend instructional time working with students to overcome limitations in design or artistic ability while also avoiding the use of copyrighted material. Students are encouraged to generate designs that resonate with the audience through whatever means necessary, including by drawing freehand or finding online images to guide their thinking. For example, some students used software programs available on their tablets to trace the shape of online images. Then, they incorporated the images directly into their book designs (and cited them appropriately in APA format). If electronic devices are not available, students can also draw their books by hand, still focusing on the characteristics of technical communication and translation of complex ideas.

I also use one additional resource to facilitate the book assignment workdays. Students are presented with several titles from the *Baby University* series by Chris Ferrie. The books demonstrate complex technical content for adolescents and toddlers (e.g. evolution, organic chemistry). Students read the books in groups and make note of the rhetorical strategies used to simplify complicated concepts. The main ideas center around the use of simple imagery, inclusion of a limited number of details, incorporation of extended metaphors, and reliance on concise language. Importantly, the books complement many students' lack of design experience by showcasing how modest or basic drawings can generate broader understanding. The book analysis could also be used to teach the rhetorical situation on day 1.

Days 7–8: Presentations

The unit concludes with the presentations to the local community at the library. To ease the burden on library partners, I split students into groups so that half attended on one day and half attended on another day. The presentation schedule also aligned with the typical hours the library offered STEM-related content to the public as part of an after-school collaboration (5:00 p.m. to 7:00 p.m.). Students are instructed to bring tablets or laptops so they can share their books with interested attendees. Moreover, to increase engagement, students prepare a brief activity that interactively demonstrates their topics. For example, one student brought several plastic pipettes to mimic the process used to study cells. Another student, whose book discussed airplanes and wind resistance, helped attendees construct paper airplanes. The book topics were shared with library partners prior to the presentations. However, students chose their own

activities. As noted, the details of the presentations were established in consultation with the library partners, who were keenly aware of their audience's needs. Thus, they are likely to change, depending on an individual's context or location when using this activity. Each session lasted approximately two hours, and students engaged with both children and their parents, who were also in attendance.

Debriefing

After the library presentations, students had three opportunities to reflect on their learning. First, I held a discussion concerning the successes and hardships with the books and presentations. Students reflected on the differences experienced when trying to explain the same complex topic across different modalities: writing and speaking (e.g. What were the differences you noticed between writing about your topic versus trying to explain it orally to your audience?). Second, students were already required to complete a six- to eight-minute speech as a graded activity in the course. As an option for this speech, they could discuss their experiences with the children's book and subsequent presentation (other options included discussing a social issue that interested the student or discussing the subject of their book—e.g. How does bitcoin work?). Although the speech was a separate assignment that did not require the public-facing experience, students who chose this option were challenged to take their experiences and turn them into engaging narratives that would be meaningful to their peers. Third, the library presentation served as the culmination of required service-learning hours in the course. Students created podcasts throughout the semester that allowed them to reflect on the impact of their service experiences. As part of the final podcast episode, many students reflected on how their library presentations impacted the community. The podcasts gave students the freedom to reflect further on how the assignment helped them develop new communication skills. However, the key takeaway is that the options facilitated reflection in multiple ways. In each case, students considered how to make their communication audience centered rather than reader, speaker, or writer centered. Technical communication depends on one's ability to think about another's needs so that problems can be anticipated and solved.

Appraisal

Students reported enjoying the library presentation portion of the activity. The opportunity to showcase their knowledge publicly was an exciting way to practice explaining technical concepts. The difficulty in this activity came from having students write their own children's literature. Although working with children's literature is not new (see Penrose, 2023), final submissions varied widely. Three themes surfaced in terms of difficulties: (1) conciseness, (2) artistic ability, and (3) audience recognition.

Students effectively applied the characteristics of technical writing needed for problem solving in STEM contexts, including explicitness, specificity, and accessibility. The best examples used metaphors or extended comparisons to put their complex topics in terms that resonated with the intended audience. However, concisely articulating details remained a struggle. Many books contained long paragraphs littered with jargon, as students struggled to shrink their subject matter

expertise into smaller chunks. Students also varied widely in artistic ability. Some students' imaginative use of tablets to copy and trace images proved helpful, but many suffered from an inability to construct complementary imagery. Future interactions of this activity might include students inserting photographs of their topics or concepts, perhaps as part of their development of the interactive activity. This would eliminate the need for drawing capabilities and offer realistic depictions of their complex phenomena.

Finally, students expressed confusion about the intended audience throughout the unit—namely, a lack of clarity surrounding the groups that would attend the library presentations inhibited students' designs. Although the *Baby University* series provided insight into concepts important for translating complex information and alleviated concerns about drawing ability, the context differed from the target audience of their presentations (which was significantly older). As one student stated, "All examples of children's books shown to us were part of a series called *Baby University*, which has a target audience significantly younger than the target of this project." Instructors might consider similar books or series that achieve the same translational goals but are targeted toward late elementary or early middle school students instead. For example, *Atoms Under the Floorboards: The Surprising Science Hidden in Your Home* by Chris Woodford or *How to Code a Sandcastle* by Josh Funk demonstrate the key characteristic of using relatable metaphors to enhance understanding of STEM concepts but are marketed toward an older audience.

Apart from the nature of the activity, two additional challenges related to community partnerships stand out. First, partnerships are not limited to libraries. The success of the activity unquestionably depends on the ability to facilitate community partnerships, but they can occur within any setting where students practice public-facing communication. This could include STEM nights at a local elementary school or community-based nonprofits that focus on STEM or engineering education. Ultimately, the instructor must make this decision, given their knowledge of the activity, local community setting, and ability to work collaboratively during the semester. Second, many classes may include nontraditional students who have outside responsibilities that make attending community functions difficult. This could potentially be navigated through the dates and times identified by the instructor and the relevant community organizations. For example, the current community event took place between 5:00 p.m. and 7:00 p.m. on two separate days near the end of the semester. It was also scheduled prior to the beginning of the semester, giving students flexibility for navigating attendance outside the classroom.

In conclusion, this activity gave students multiple opportunities to understand effective technical communication better. By working with the public, students applied their knowledge in an impact way and became better prepared to communicate their technical expertise when they eventually enter the workforce.

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